

Private DP teacher training stage

Sensitive private speech data $\mathcal{D}_{\text{priv}}$



imbalanced class distribution

Frozen speech encoder + classifier head

(ϵ, δ) -DP w.r.t. $\mathcal{D}_{\text{priv}}$

DP-SGD

per-example clipping + Gaussian noise

DP teacher θ_t

DP protects private training records.

Teacher health diagnosis on $\mathcal{D}_{\text{dev}}$

DP teacher θ_t

Development set $\mathcal{D}_{\text{dev}}$

Confusion matrix
Balanced accuracy
Macro-F1
Majority prediction rate
Per-class recall

Global health $\hat{H}$
Class health $\hat{h}_k$
Uncertainty $\hat{\sigma}_H$
✓ healthy teacher
⚠ collapsed teacher

Permitted auxiliary supervision

Auxiliary speech data $\mathcal{D}_{\text{aux}}$

Optional offline teacher query

Teacher logits $Q_{\theta_t}(\mathcal{D}_{\text{aux}})$

Independent hard labels y_{aux}

Teacher is not released and not deployed as a query interface.

Teacher-Conditioned Release Distillation TCRD

low-health $\hat{H} < \tau_{\text{low}}$

Hard-label training
 $L_{\text{hard}} = \text{CE}(z_s, y)$
Suppress unreliable teacher transfer.

intermediate-health $\tau_{\text{low}} \leq \hat{H} < \tau_{\text{high}}$

Health-aware KD
 L_{HAKD}

Weight teacher supervision by class health $\hat{h}_k$.

high-health $\hat{H} \geq \tau_{\text{high}}$

Standard KD
 L_{KD}

Use soft targets when teacher preserves balanced structure.

Health-guided routing using $\hat{H}$, τ_{low} , and τ_{high}

Train student on $\mathcal{D}_{\text{aux}}$

Released audio-only student θ_s